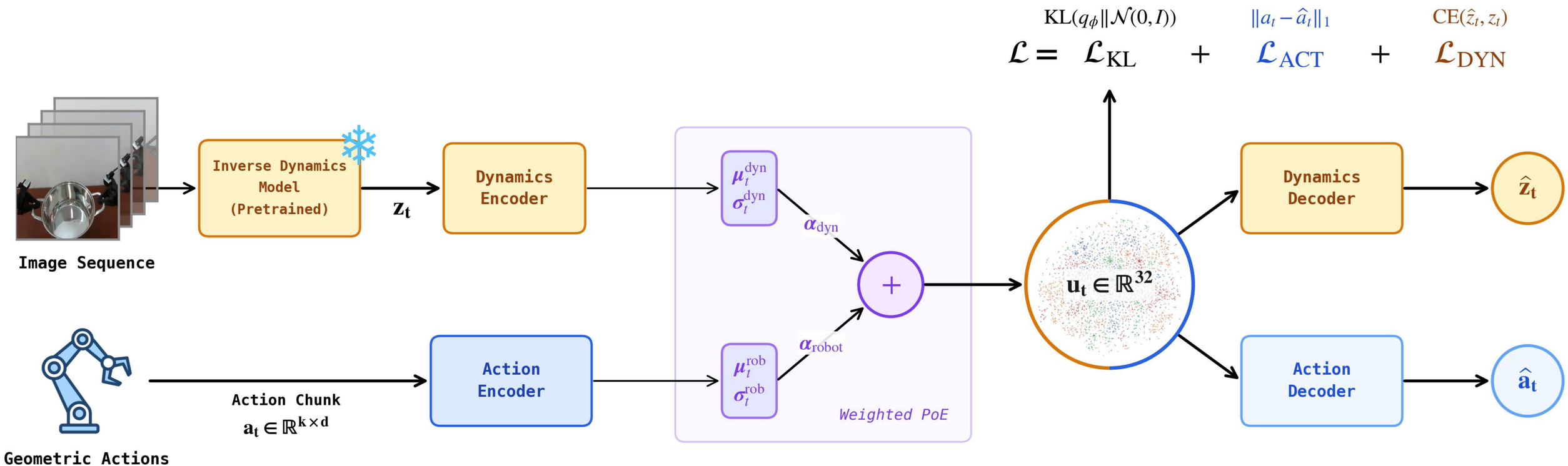
$\mathrm{KL}(q_\phi \| \mathcal{N}(0,I))$
$\|a_t - \hat{a}_t\|_1$
$\mathrm{CE}(\hat{z}_t, z_t)$
$\mathcal{L} = \mathcal{L}_{\mathrm{KL}} + \mathcal{L}_{\mathrm{ACT}} + \mathcal{L}_{\mathrm{DYN}}$
Image Sequence
Inverse Dynamics Model (Pretrained)
z_t
Dynamics Encoder
μ_t^{dyn}
σ_t^{dyn}
α_{dyn}
+
α_{robot}
μ_t^{rob}
σ_t^{rob}
Weighted PoE
$u_t \in \mathbb{R}^{32}$
Dynamics Decoder
$\hat{z}_t$
Action Decoder
$\hat{a}_t$
Geometric Actions
Action Chunk
$a_t \in \mathbb{R}^{k \times d}$
Action Encoder